

into the different tasks associated with the phases of the development process. Either way, once the overall structure is determined, tasks are identified and included in the WBS. We return to the topic of WBSs and their use in iterative planning later in this chapter.

Gantt Chart

A *Gantt chart* is a horizontal bar chart that shows the same task information as the project WBS but in a graphical way. Sometimes a picture really is worth a thousand words, and the Gantt chart can communicate the high-level status of a project much faster and easier than the WBS. Creating a Gantt chart is simple and can be done using a spreadsheet package, graphics software, or a project management package.

First, tasks are listed as rows in the chart, and time is listed across the top in increments based on the needs of the projects (see Figure 2-13). A short project may be divided into

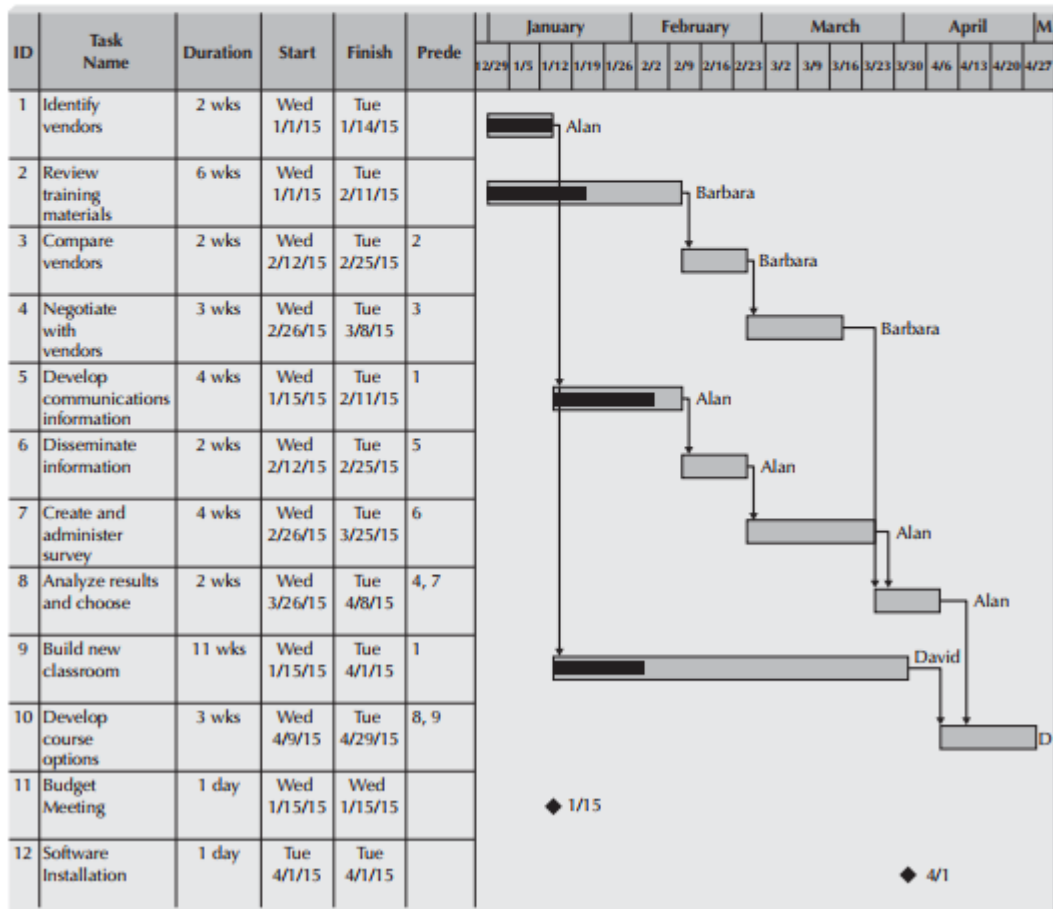


FIGURE 2-13 Gantt Chart

development also requires an incremental and iterative process. In the description of the enhanced Unified Process in Chapter 1, the development process was organized around iterations, phases, and workflows. In many ways, a workplan for an incremental and iterative development process is organized in a similar manner. For each iteration, there are different tasks executed on each workflow. This section describes an incremental and iterative process using evolutionary WBSs for project planning that can be used with object-oriented systems development.

Evolutionary WBSs allow the analyst to develop an *iterative workplan*. First, evolutionary WBSs are organized in a standard manner across all projects: by workflows, phases, and then the specific tasks that are accomplished during an individual iteration. Second, evolutionary WBSs are created in an incremental and iterative manner. This encourages a more realistic view of both cost and schedule estimation. Third, because the structure of an evolutionary WBS is not tied to any specific project, evolutionary WBSs enable the comparison of the current project to earlier projects. This supports learning from past successes and failures.

In the case of the enhanced Unified Process, the workflows are the major points listed in the WBS. Next, each workflow is decomposed along the phases of the enhanced Unified Process. After that, each phase is decomposed along the tasks that are to be completed to create the deliverables associated with an individual iteration contained in each phase (see Figure 1-16). The template for the first two levels of an evolutionary WBS for the enhanced Unified Process would look like Figure 2-17.

As each iteration through the development process is completed, additional iterations and tasks are added to the WBS (i.e., the WBS evolves along with the evolving information system).⁸

I. Business Modeling	V. Implementation	IX. Project Management
a. Inception	a. Inception	a. Inception
b. Elaboration	b. Elaboration	b. Elaboration
c. Construction	c. Construction	c. Construction
d. Transition	d. Transition	d. Transition
e. Production	e. Production	e. Production
II. Requirements	VI. Test	X. Environment
a. Inception	a. Inception	a. Inception
b. Elaboration	b. Elaboration	b. Elaboration
c. Construction	c. Construction	c. Construction
d. Transition	d. Transition	d. Transition
e. Production	e. Production	e. Production
III. Analysis	VII. Deployment	XI. Operations and Support
a. Inception	a. Inception	a. Inception
b. Elaboration	b. Elaboration	b. Elaboration
c. Construction	c. Construction	c. Construction
d. Transition	d. Transition	d. Transition
e. Production	e. Production	e. Production
IV. Design	VIII. Configuration and Change Management	XII. Infrastructure Management
a. Inception	a. Inception	a. Inception
b. Elaboration	b. Elaboration	b. Elaboration
c. Construction	c. Construction	c. Construction
d. Transition	d. Transition	d. Transition
e. Production	e. Production	e. Production

FIGURE 2-17
Evolutionary WBS
Template for the
Enhanced Unified
Process